

PAM windows workstations

Dan Woodrich

Attributes

- Windows instances on GCP
- Flexible hardware: self-service possible while informing us of changes
- We use a series of images for software delivery- meaning to use the latest standard package, we support with a delete/create
- Built-in features
 - Mounts to PAM GCS buckets to allow for desktop use of GCS data
 - PAM software (raven, triton, pamguard, etc)
 - IDL license server integration
 - User mount / read-ahead cache software

Challenges

- “Reading” an acoustic file is now under the hood, a download.
 - Time to first byte latency - makes rapid reads of small files much slower
 - Download speed of large files: can be limited by network speeds
- Mounts are fragile, finicky software. When they have issues, it’s also not obvious to users what is wrong
- NMFS hardening (security policies) can cause lots of hard to track down issues with applications
- High costs of Windows licenses

Mitigations

- Commissioned user mount / read ahead cache tool to speed up file retrieval over mounts
- Created test branch for experimenting with relaxing security rules for easier assessment and security approval
- Identify heavier processing use cases and attempt to support in cloud native processing

Demo

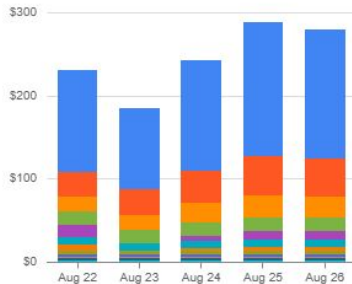
- Main features of pam-ww
 - Login (user and admin)
 - Mounts (default and user)
 - Software
- Users commonly ask: what's the right way to use the user mount / read ahead cache tool when working over mounts for interactive workloads, and what kind of performance can I expect?

Fleet

- 75 instances
- All NMFS science centers represented, plus a few other guests from other SIs
- Cost breakdown

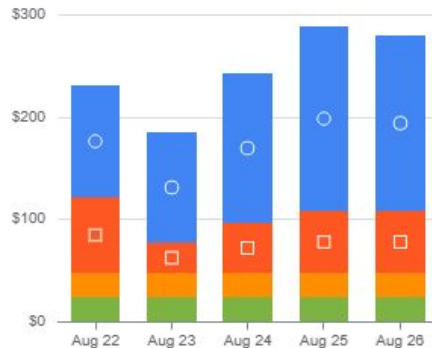
Dominated by windows license costs.

For apps that need windows: much cheaper than software teams



SKU

- Licensing Fee for Windows Server 2022 Data (license!)
- E2 Instance Core running in Virginia
- ◆ E2 Instance Ram running in Virginia
- ▼ Storage PD Capacity in Virginia



product_name

- product_name:pam-ww (normal use)
- product_name:pam-ww-extension (higher compute use)
- ◆ [Charges for other usage] (non pam-ww, like cloud native)
- ▼ product_name:pam-ww-gpu (gpu use)

Next steps

- NMFS windows workstations: may absorb pam-ww at some point
 - Carve off more and more processing use cases to cloud native processing
 - Eventually: interactive use cases go to web applications that are GCS / makara aware, for natural use of data and extension to NMFS
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- VMs on cloud will always persist in some form: development and analysis, one-off scientific use (casual R/matlab/python analysis). Linux is a great option if possible due to much lower costs.